Homology Modeling and Molecular Docking Analysis of TviB from Salmonella enterica serovar Typhi

A Computational Structural Biology Mini-Project

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Abstract

Typhoid fever is caused by the bacteria Salmonella Typhi, which protects itself from the human immune system using a protective outer coat called Vi capsule. An enzyme called TviB performs the first step in building this coat, making it a useful protein to study — but no one has ever determined its actual 3D shape in the lab. This project built a predicted 3D model of TviB using a computational method called homology modeling, borrowing the shape of a similar, already-solved bacterial enzyme (Cap5O, 31% sequence similarity, usable) as a template. The model was checked for reliability using standard quality tests, scoring 0.67 (GMQE) and 0.66 ± 0.05 (QMEAN), with 91.85% of the structure falling into physically realistic shapes on a Ramachandran plot. The model was then tested by fitting TviB's natural partner molecule, UDP-GlcNAc, into it using molecular docking, which returned a favourable binding score of −9.1 kcal/mol. To confirm this result was biologically meaningful and not a coincidence, the amino acids involved in the fit were compared against amino acids already confirmed, in the template's real structure, to be important for binding its own partner molecule. Of 20 amino acids checked, 13 matched across four separate regions of the protein, providing good evidence that the model's active site is realistic. This project did not produce a drug for typhoid; instead, it built and validated a trustworthy structural starting point for TviB, which could support future, more advanced work on this understudied protein.

1. Introduction

1.1 Biological Background

Typhoid fever, caused by Salmonella enterica serovar Typhi, remains a significant public health burden in many developing regions. A key virulence factor of S. Typhi is the Vi capsular polysaccharide, a linear polymer of α-1,4-linked N-acetylgalactosaminuronic acid that shields the bacterium from host immune recognition and forms the basis of the licensed Vi polysaccharide vaccine. Biosynthesis of the Vi antigen is directed by the viaB locus, which encodes nine genes:

- tviB, tviD, tviC, tviE – build actual sugar chains that makes vi capsule and
- vexA, vexB, vexC, vexD, vexE – handles exporting the materials to make vi capsule)

TviB and TviC catalyse the two committed enzymatic steps that generate the activated nucleotide-sugar precursor for Vi polymer assembly. TviB has been proposed as a candidate target for anti-virulence therapeutics and as a diagnostic marker specific to S. Typhi because from this pathway begins, blocking the first step is better. Secondly, experimented kinetic data of TviB is present in databases.

1.2 Rationale for Homology Modeling

A search of the RCSB Protein Data Bank confirmed that no experimentally determined structure exists for TviB from S. Typhi. This absence of structural data is the central justification for the present study: without a solved structure, structure-based investigation

of TviB — including docking studies to evaluate substrate binding or to support future inhibitor design — is not possible without first generating a reliable homology model.

1.3 Objectives

- Identify a suitable structural template for TviB via sequence similarity search (BLAST) against the PDB.
- Construct a homology model of TviB using SWISS-MODEL.
- Validation of the model using complementary metrics (GMQE, QMEAN, MolProbity, Ramachandran analysis).
- Dock the native substrate, UDP-GlcNAc, into the modeled active site using AutoDock Vina.
- Cross-validate the docking result against the template's experimentally known cofactor (NAD)-binding residues via sequence alignment, to assess whether the predicted binding site is biologically consistent.

2. Methodology

2.1 Sequence Retrieval

The TviB amino acid sequence (425 residues) was retrieved from UniProt for Salmonella enterica serovar Typhi.

2.2 Template Identification

The TviB sequence was submitted to protein BLAST (blastp) against the PDB database. The top-scoring hit was **Chain A of Cap5O from Staphylococcus aureus (PDB ID: 3OJL), a UDP-glucose/UDP-GlcNAc dehydrogenase** family enzyme, with 31.02% sequence identity (usable), 98% query coverage, and an E-value of 1e–53. All other hits returned identities below 31%, confirming 3OJL as the best available template. The template structure was independently examined on RCSB PDB: it was solved by X-ray diffraction at 2.80 Å resolution, assembles as a homodimer, and is co-crystallised with the cofactor NAD.

2.3 Homology Model Construction

The TviB sequence was submitted to SWISS-MODEL with 3OJL (Chain A) manually selected as the modelling template, in preference to automated template selection, to ensure the modelling basis matched the independently verified BLAST result.

2.4 Model Validation

Model quality was assessed using two complementary approaches. First, SWISS-MODEL's own scoring metrics — Global Model Quality Estimate (GMQE) and QMEANDisCo — were recorded, which assess overall model reliability relative to the target–template alignment.

Second, stereochemical quality was independently assessed using MolProbity, examining the Ramachandran plot distribution, clash score, and overall MolProbity score.

2.5 Ligand Preparation

UDP-N-acetylglucosamine (UDP-GlcNAc), the native substrate of TviB, was selected as the docking ligand on the basis of its experimentally confirmed enzymatic role. Its three-dimensional structure (SDF, 3D conformer) was obtained from PubChem.

2.6 Protein and Ligand Preparation for Docking

Since, the biological assembly of the TviB–Cap5O template is a homodimer, and SWISS-MODEL's dimeric prediction ("matching prediction") was noted, Chain B alone was extracted in PyMOL for docking to maintain a tractable single-active-site system; this simplification is discussed as a limitation in Section 5. Both chains carry an identical sequence, so this choice does not affect the docking chemistry. Solvent molecules were removed, polar hydrogens were added, and the structure was exported for conversion to PDBQT format using AutoDockTools (MGLTools). The ligand was similarly processed, with Kollman charges and rotatable bonds assigned automatically.

2.7 Active Site Definition and Docking

As the SWISS-MODEL structure did not retain the template-bound NAD (an expected outcome, since ligand transfer was not requested during modeling), the docking grid box was centred on the region homologous to the template's known NAD-binding loops, informed by residue-level correspondence established via pairwise sequence alignment (Section 2.8). Molecular docking was performed using AutoDock Vina.

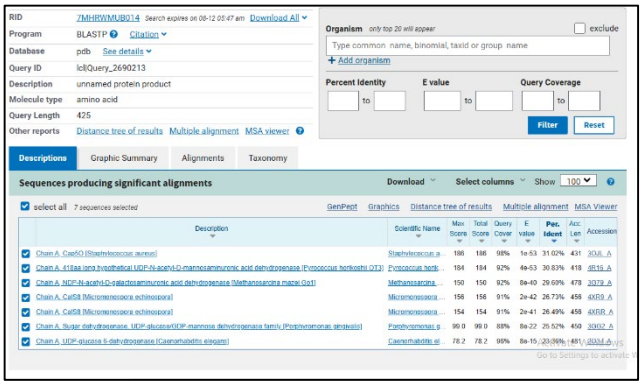
2.8 Active Site Cross-Validation via Sequence Alignment

To assess whether the docked pose occupied a biologically meaningful site, residues within 4 Å of the top-ranked docking pose were identified in PyMOL. Independently, residues contacting the bound NAD cofactor in the 3OJL template were retrieved from the RCSB Ligand Interaction viewer. Because template (Cap5O) and target (TviB) residue numbering are not equivalent — a consequence of insertions, deletions, and an N-terminal purification tag in the template construct — a pairwise global alignment (EMBOSS Needle, 30.1% identity, 462 aligned positions) was used to map each NAD-contact residue in Cap5O to its structurally corresponding position in the TviB model. This mapped list was then directly compared against the docking-derived active-site residue list.

3. Results

3.1 Template Selection

3OJL (Cap5O, S. aureus) was selected as template with 31.02% sequence identity and 98% query coverage — above the ~30% identity threshold generally considered usable for homology modeling, though at the lower end of that range.



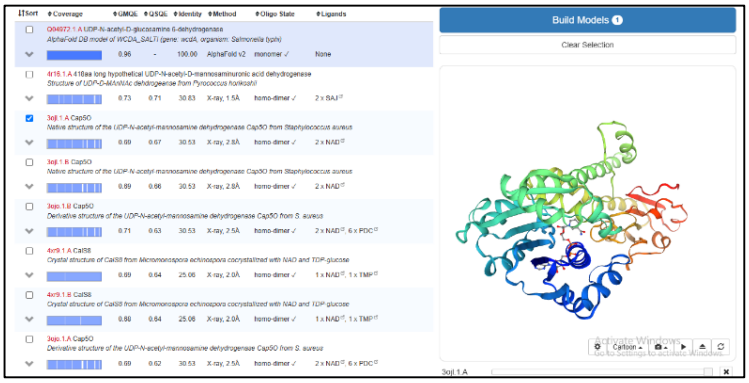


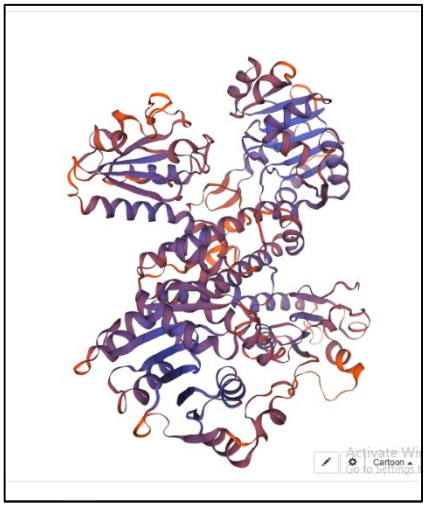
Fig1(a & b): Blast results (Cap5O shows 31.02% sequence identity)

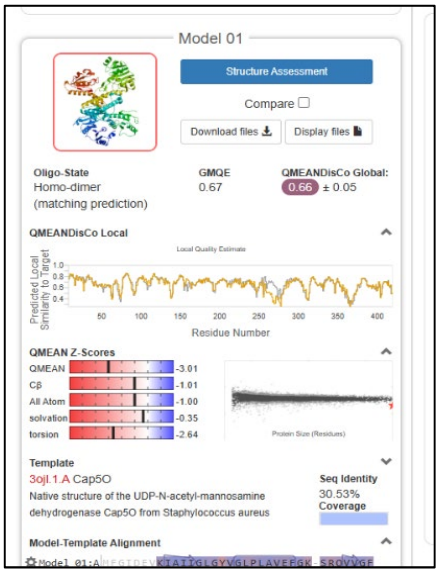
3.2 Model Quality

Table 1 summarises the validation metrics obtained for the TviB homology model.

Metric	Score	Interpretation
Template identity (3OJL)	31.02%	Borderline but usable
Template resolution	2.80 Å	Moderate quality
GMQE	0.67	Moderate confidence
QMEANDisCo Global	0.66 ± 0.05	Acceptable (>0.6 threshold)
QMEAN Z-score	−3.01	Below ideal (0); reflects template limitations
MolProbity score	1.85	Good (lower is better)
Clash score	1.14	Very good
Ramachandran favoured	91.85%	Passes 90% threshold
Ramachandran outliers	2.07%	Acceptable (<5%)

Table 1. Homology model validation metrics for the TviB model built on the 3OJL template.





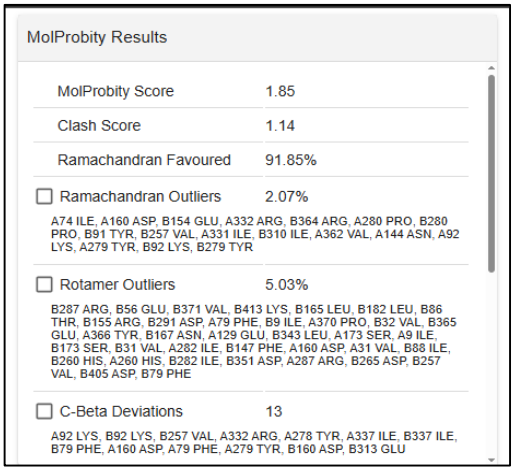


Fig 2a: TviB structure | Fig 2b: QMEANisCo Global and QMEAN Z-score | Fig 2c: MolProbity Results

The model's GMQE (0.67) and QMEAN Z-score (−3.01) reflect the moderate template identity and indicate the model should be treated as a moderate-confidence prediction rather than a high-accuracy structure. In contrast, MolProbity-based stereochemical validation was notably stronger, with 91.85% of residues in favoured Ramachandran regions and a low clash score (1.14), indicating that the model's local geometry is physically reasonable even where global fold confidence is only moderate. Both sets of scores are reported here without selective omission, since they measure different properties of the model and neither alone is sufficient to judge its usability.

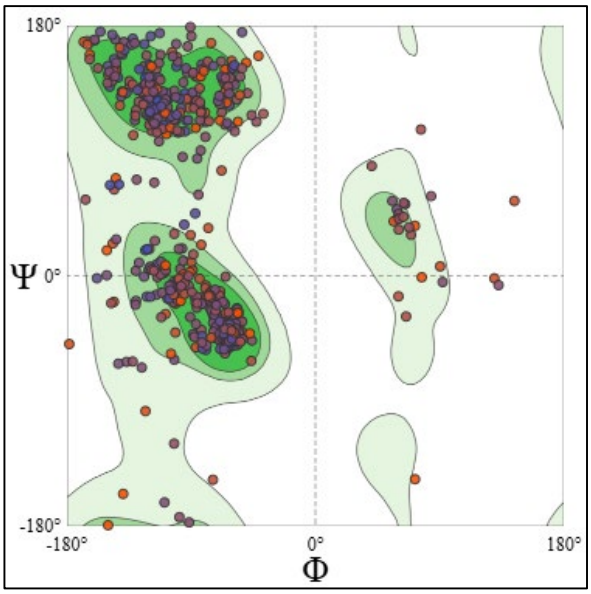


Fig 3: Ramachandran Plot results

3.3 Docking Results

UDP-GlcNAc was docked into the TviB homology model (Chain B) using AutoDock Vina. The top-ranked pose returned a predicted binding affinity of −9.104 kcal/mol, indicating a favourable predicted interaction.

mode	affinity (kcal/mol)	dist from best mode rmsd l.b.	rmsd u.b.
1	-9.104	0	0
2	-8.861	2.647	3.593
3	-8.834	2.689	7.247
4	-8.784	2.514	8.096
5	-8.766	2.9	8.535
6	-8.765	1.861	2.928
7	-8.733	2.848	7.846
8	-8.727	2.802	5.401
9	-8.653	2.878	5.956
10	-8.637	3.386	5.82
11	-8.634	2.883	8.255
12	-8.627	2.466	7.936
13	-8.597	1.739	2.949
14	-8.524	2.746	7.141
15	-8.502	2.346	6.857
16	-8.497	2.458	7.947
17	-8.492	2.521	4.493
18	-8.487	2.809	7.991
19	-8.485	2.366	4.617
20	-8.437	2.464	7.056

Fig 4: Binding Affinity Results

3.4 Active Site Cross-Validation

Residues within 4 Å of the top-ranked docked pose in the TviB model were identified as: 15, 16, 17, 83, 84, 85, 86, 120, 121, 154, 155, 156, 208, 211, 212, 215, 246, 259, 260, and 261. Pairwise alignment of TviB against the 3OJL template (Table 2) was used to map the template's experimentally confirmed NAD-contact residues onto the corresponding TviB positions.

Thirteen of the twenty active-site residues identified from the docking pose (65%) correspond to confirmed NAD-binding positions in the template, distributed across four independent, non-contiguous sequence clusters (residues 15–17, 83–86, 120–121, and 154–156, plus Cys-261). Several of these correspondences are exact residue-type matches (e.g., Glu→Glu, Arg→Arg, Ser→Ser, Thr→Thr, Cys→Cys), suggesting the model has retained functionally relevant side chains at structurally conserved positions rather than the overlap being coincidental. The remaining five residues (208, 211, 212, 215, 246) did not correspond to confirmed NAD-contact positions; since UDP-GlcNAc is a larger substrate that occupies a

binding surface adjacent to, but distinct from, the NAD cofactor pocket, this is consistent with expected substrate-specific contacts rather than indicating an erroneous docking result.

TviB residue	Corresponding 3OJL (Cap5O) residue	Confirmed NAD contact?
Gly-15	Gly-9	Yes
Tyr-16	Tyr-10	Yes
Val-17	Ile-11	Yes
Thr-83	Ala-79	Yes
Val-84	Val-80	Yes
Pro-85	Pro-81	Yes
Thr-86	Thr-82	Yes
Ser-120	Ser-118	Yes
Thr-121	Thr-119	Yes
Glu-154	Glu-151	Yes
Arg-155	Arg-152	Yes
Ile-156	Val-153	Yes
Cys-261	Cys-258	Yes
Lys-208 / Glu-211 / Asn-212 / Arg-215 / Gly-246	(nearby, non-NAD positions)	No

Table 2. Mapping of docking-derived active-site residues to template NAD-contact positions via pairwise sequence alignment.

3.5 Docking Pose Visualisation

[Insert ray-traced PyMOL figure here: TviB model shown as cartoon, docked UDP-GlcNAc shown as sticks, matched active-site residues (Gly-15, Tyr-16, Val-17, Thr-83, Val-84, Pro-85, Thr-86, Ser-120, Thr-121, Glu-154, Arg-155, Ile-156, Cys-261) shown as labelled sticks.]

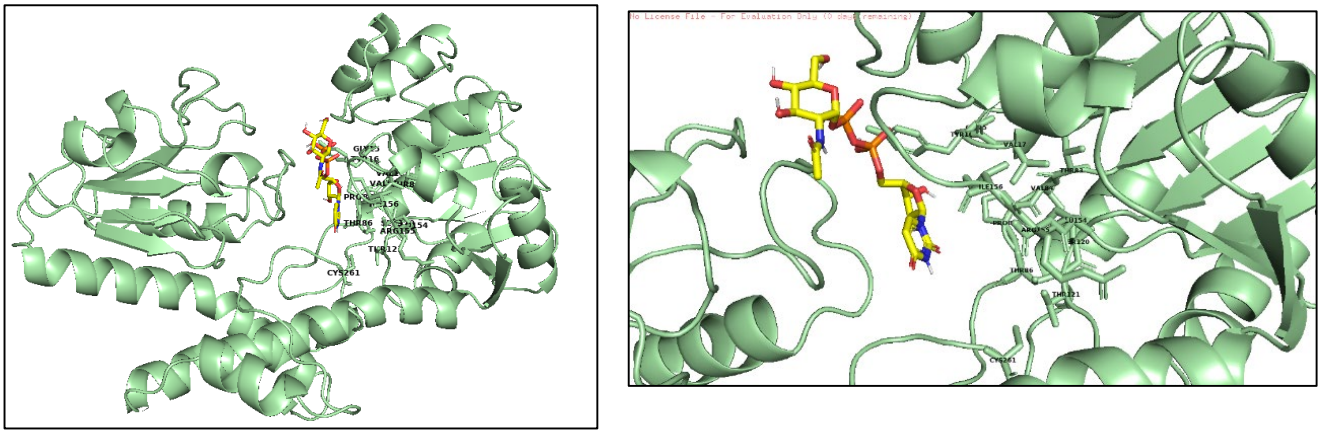


Fig 5: Docked pose of UDP-GlcNAc within the TviB homology model active site, with cross-validated residues labelled.

4. Discussion

This study demonstrates a complete, internally validated workflow for investigating a structurally uncharacterised but biologically and clinically relevant enzyme. In the absence of an experimental TviB structure, homology modeling provided a usable, if moderate-confidence, structural framework. Rather than treating the docking score in isolation — a common shortcoming in docking-only analyses, since binding affinity alone does not confirm biological relevance — this study incorporated an explicit cross-validation step, mapping the docking-derived active site onto the template's experimentally defined cofactor-binding residues through sequence alignment. The substantial overlap (13/20 residues, four independent clusters) provides reasonable support that the model's predicted TviB active site occupies the expected, evolutionarily conserved region of this dehydrogenase family, rather than an arbitrary surface pocket.

The partial, rather than complete, overlap between the docking site and the NAD-contact residues is itself biologically informative. UDP-GlcNAc and NAD occupy adjacent but non-identical sub-sites within the active clefts of NAD^+-dependent sugar dehydrogenases; the five non-overlapping residues identified here (208, 211, 212, 215, 246) plausibly represent substrate-specific contacts distinct from the cofactor-binding loops traced from the template. This distinction was not independently confirmed against a substrate-bound template structure and should be regarded as a reasonable interpretation rather than an established finding.

5. Limitations

- Template identity was 31.02%, close to the lower boundary generally considered reliable for homology modeling; the resulting model (GMQE 0.67, QMEAN Z-score −3.01) should be regarded as moderate- rather than high-confidence.
- The template structure (3OJL) itself was solved at a moderate resolution (2.80 Å) with an R-free of 0.263, which is workable but not high-quality by crystallographic standards.
- The SWISS-MODEL structure is an apo model; the NAD cofactor present in the template was not transferred, so the active site was defined indirectly via sequence-based residue mapping rather than direct structural reference to a bound cofactor in the model itself.
- TviB's biological assembly is a homodimer, but docking was performed against a single extracted chain (monomer) for tractability. Any contribution to the active site from residues at the dimer interface would not be captured by this model.
- Active-site cross-validation was performed only against NAD-contact residues; no independent confirmation of the UDP-GlcNAc-specific sub-site (e.g., from a substrate-bound structure in a related enzyme) was available for comparison.
- Docking was performed using a single scoring function (AutoDock Vina); binding affinity values from docking are approximate free-energy estimates and were not validated against experimental binding data for TviB.

6. Conclusion

In the absence of an experimentally determined structure, a homology model of TviB was constructed using the 3OJL (Cap5O) template and validated through complementary quality metrics. Docking of the native substrate UDP-GlcNAc yielded a favourable predicted binding affinity (−9.104 kcal/mol), and independent cross-validation via sequence alignment showed that the majority of predicted active-site residues correspond to experimentally confirmed cofactor-binding positions in the template. While the moderate template identity and several structural simplifications (monomeric, apo model) limit the confidence that can be placed in this model, the workflow presented here — particularly the explicit residue-level cross-validation step — offers a more rigorous basis for interpreting the docking result than binding affinity alone, and may serve as a starting point for future structure-guided investigation of TviB as a typhoid-relevant drug target.

7. References

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